

16. Refractive index of diamond with respect to glass is 1.6. If the absolute refractive index of glass is 1.5, the absolute refractive index of diamond is :

(A) 1.09

(B) 1.96

(C) 2.4

(D) 2.5

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(C)/ 2.4

14. If the absolute refractive indices of two media X and Y are $\frac{6}{5}$ and $\frac{4}{3}$ respectively, then the refractive index of Y with respect to X will be :

(A) $\frac{10}{9}$

(B) $\frac{9}{10}$

(C) $\frac{9}{8}$

(D) $\frac{8}{9}$

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14. If the absolute refractive indices of two media X and Y are $\frac{6}{5}$ and $\frac{4}{3}$ respectively, then the refractive index of Y with respect to X will be :

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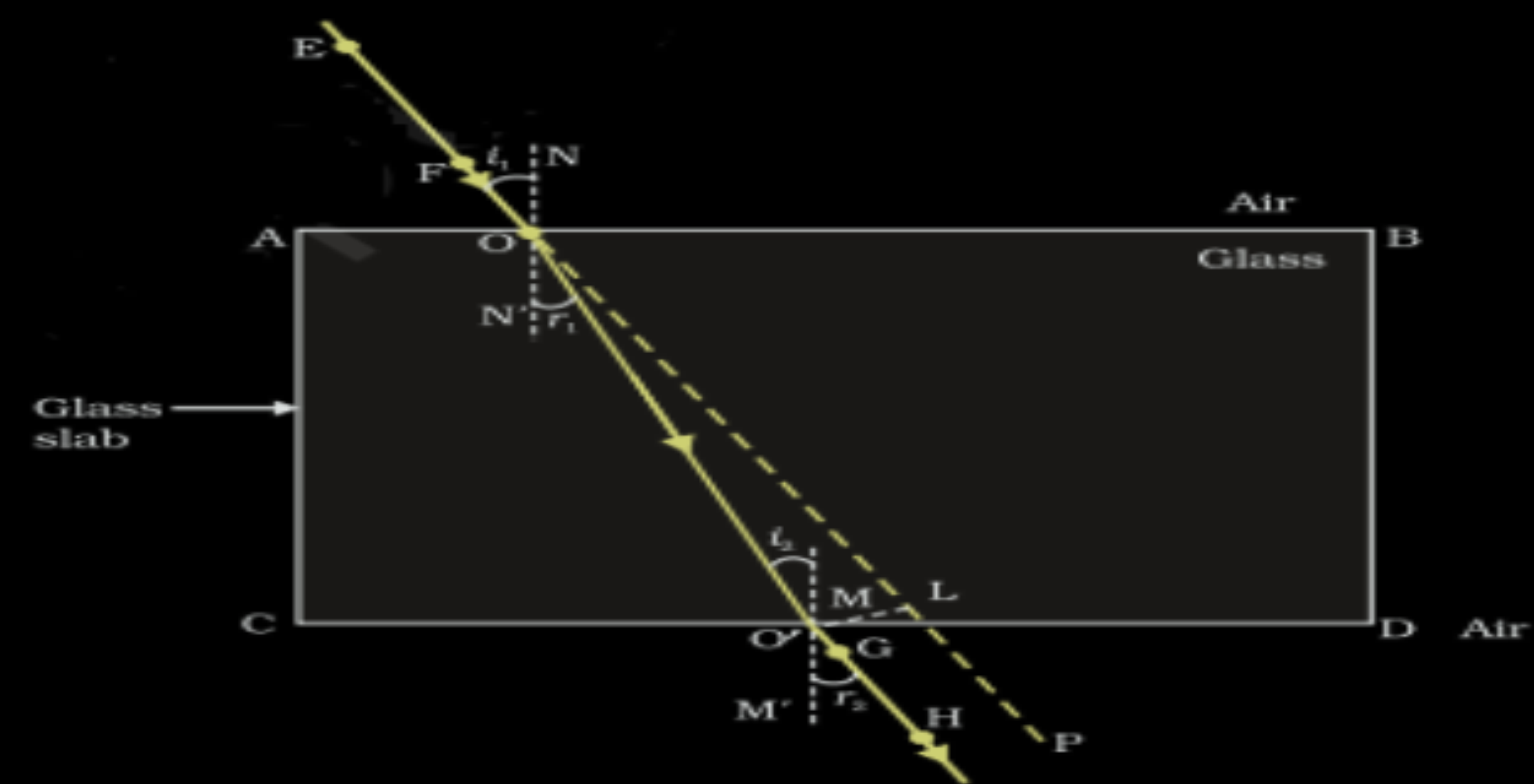
(D) $\frac{8}{9}$

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(A) / $\frac{10}{9}$

- (b) (i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.
- (ii) State Snell's law of refraction of light.
- (iii) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of :
- (I) object distance, and
- (II) magnification.

(b) (i)



(if arrows not marked , deduct half mark)

(ii) The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for a given pair of media. /

$$\frac{\sin i}{\sin r} = \text{constant}$$

(iii)

Convex Lens	Concave Lens
(I) Object to be placed between O and F	Object can be placed anywhere in front of the lens
(II) Magnified image	Diminished image

2

1

1

1

11. Absolute refractive index of water and glass is $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4}$ m/s

(B) $\frac{7}{3}$ m/s

(C) $\frac{16}{9}$ m/s

(D) $\frac{9}{8}$ m/s

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(C) $\frac{16}{9}$ m/s

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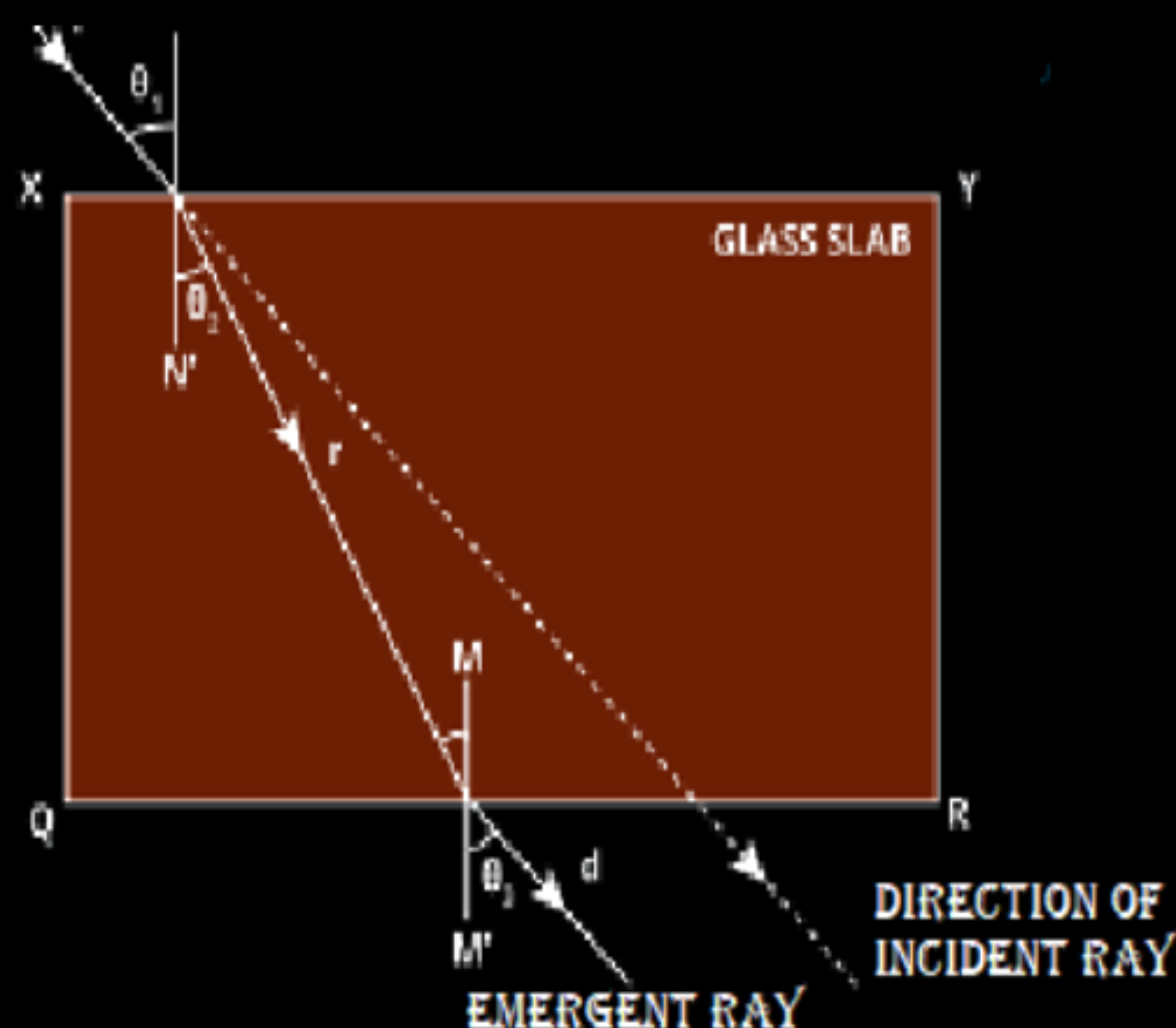
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NOTE: - Full credit of one mark to all the students.

- 36.** (a) (i) “In refraction of light through a rectangular glass slab, the emergent ray is always parallel to the direction of the incident ray.” Why ? Explain with the help of a ray diagram. What happens when a ray of light falls normally on one of the faces of a rectangular glass prism ? Draw diagram.
- (ii) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 20 cm. Use Lens formula to determine the position of the image formed in this case.

(a) (i)

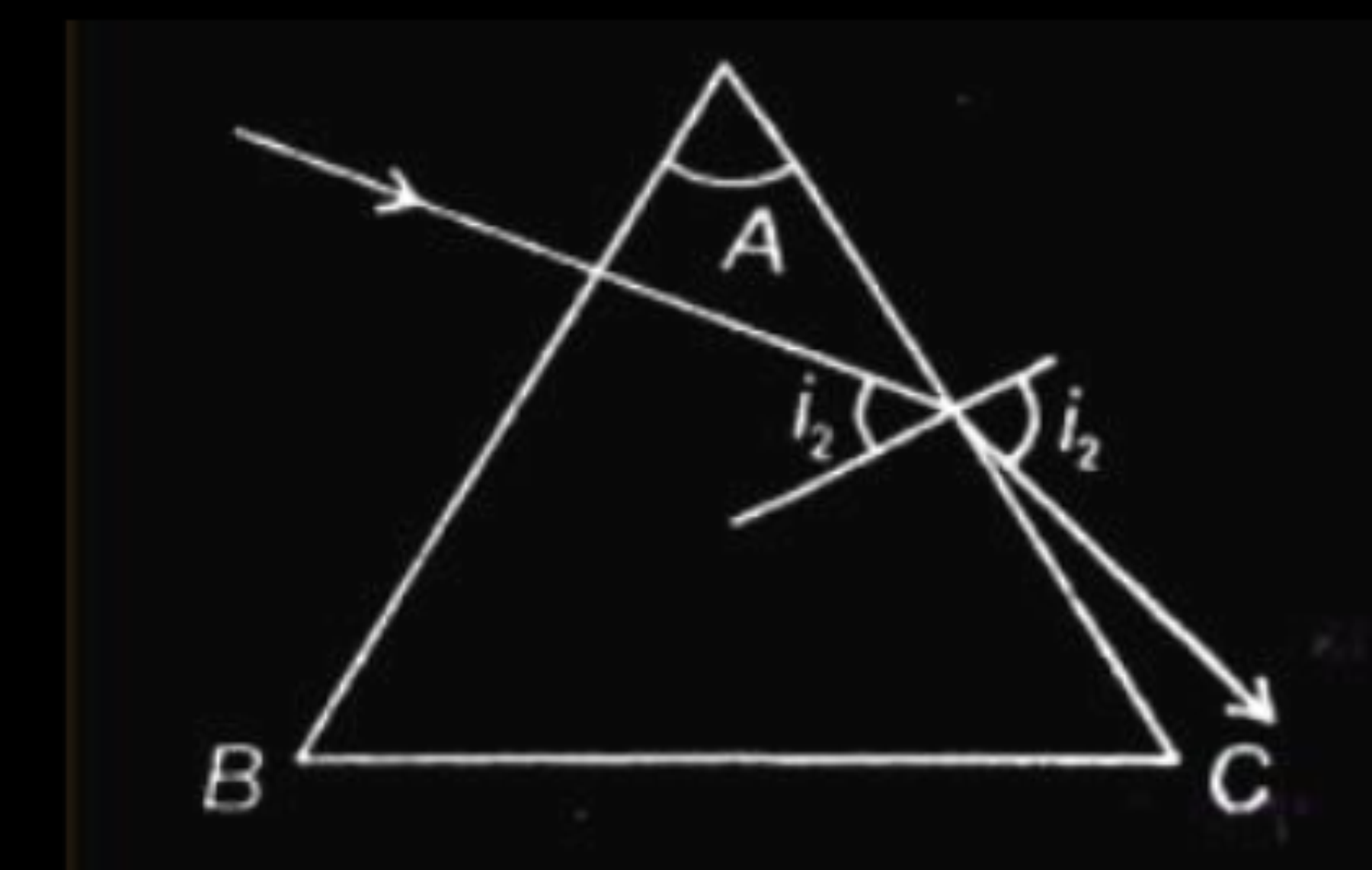
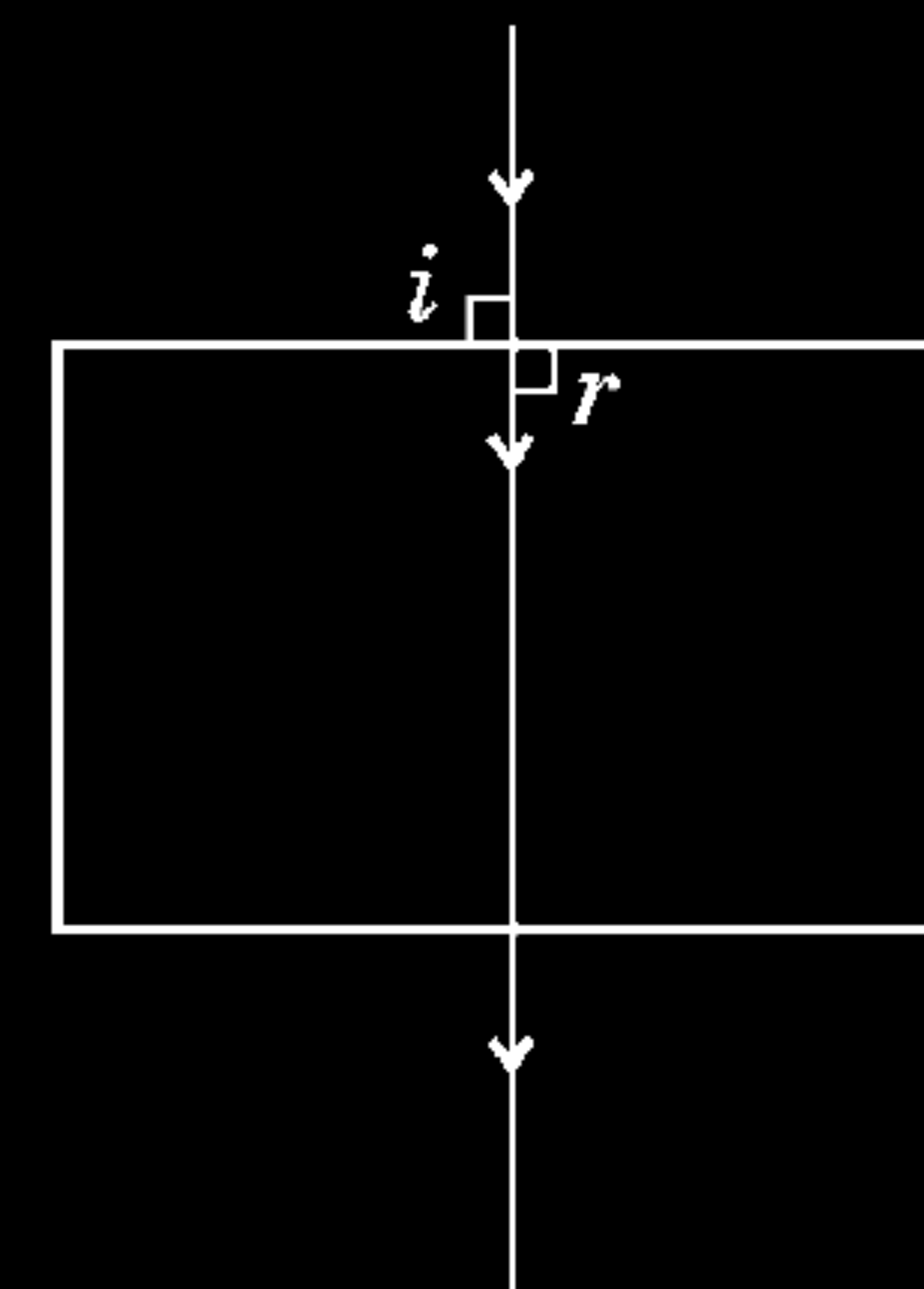
The extent of bending of the ray of light at the opposite parallel faces of the rectangular glass slab is equal and opposite. This is why the ray emerges parallel to the incident ray.



1

1

NORMAL INCIDENCE



1

(ii) $u = -30 \text{ cm}$, $f = -20 \text{ cm}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-30 \text{ cm}} + \frac{1}{-20 \text{ cm}}$$

$$= \frac{1}{-12 \text{ cm}} \quad \text{or} \quad v = -12 \text{ cm}$$

The image is at -12 cm on the same side of the lens.

$\frac{1}{2}$

$\frac{1}{2}$

1

11. An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light. If the emergent beam gets displaced laterally, the device 'X' is :

1

- (a) plane mirror
- (b) convex lens
- (c) glass slab
- (d) glass prism

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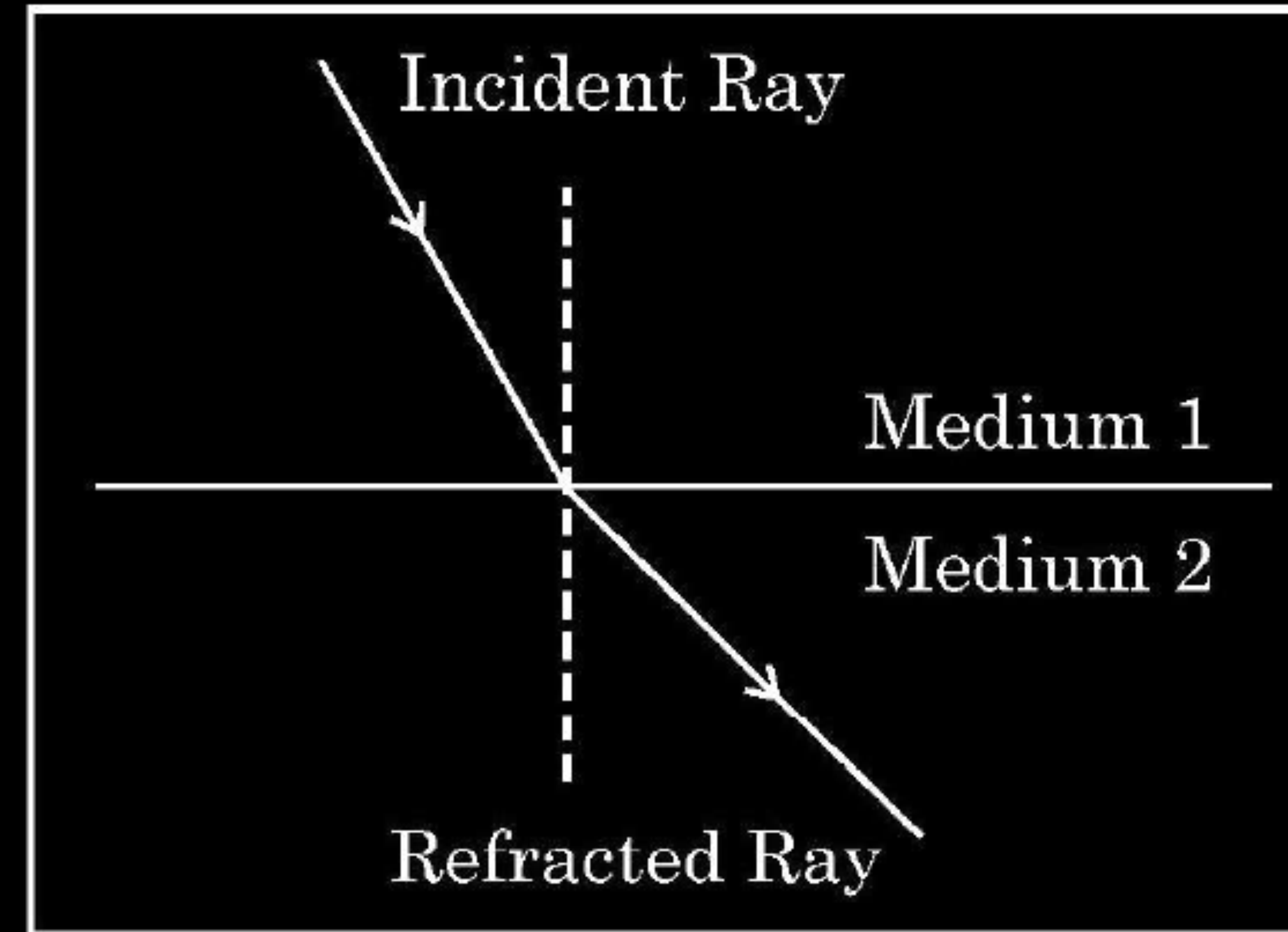
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- (a) plane mirror
- (b) convex lens
- (c) glass slab
- (d) glass prism

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(c)/glass slab

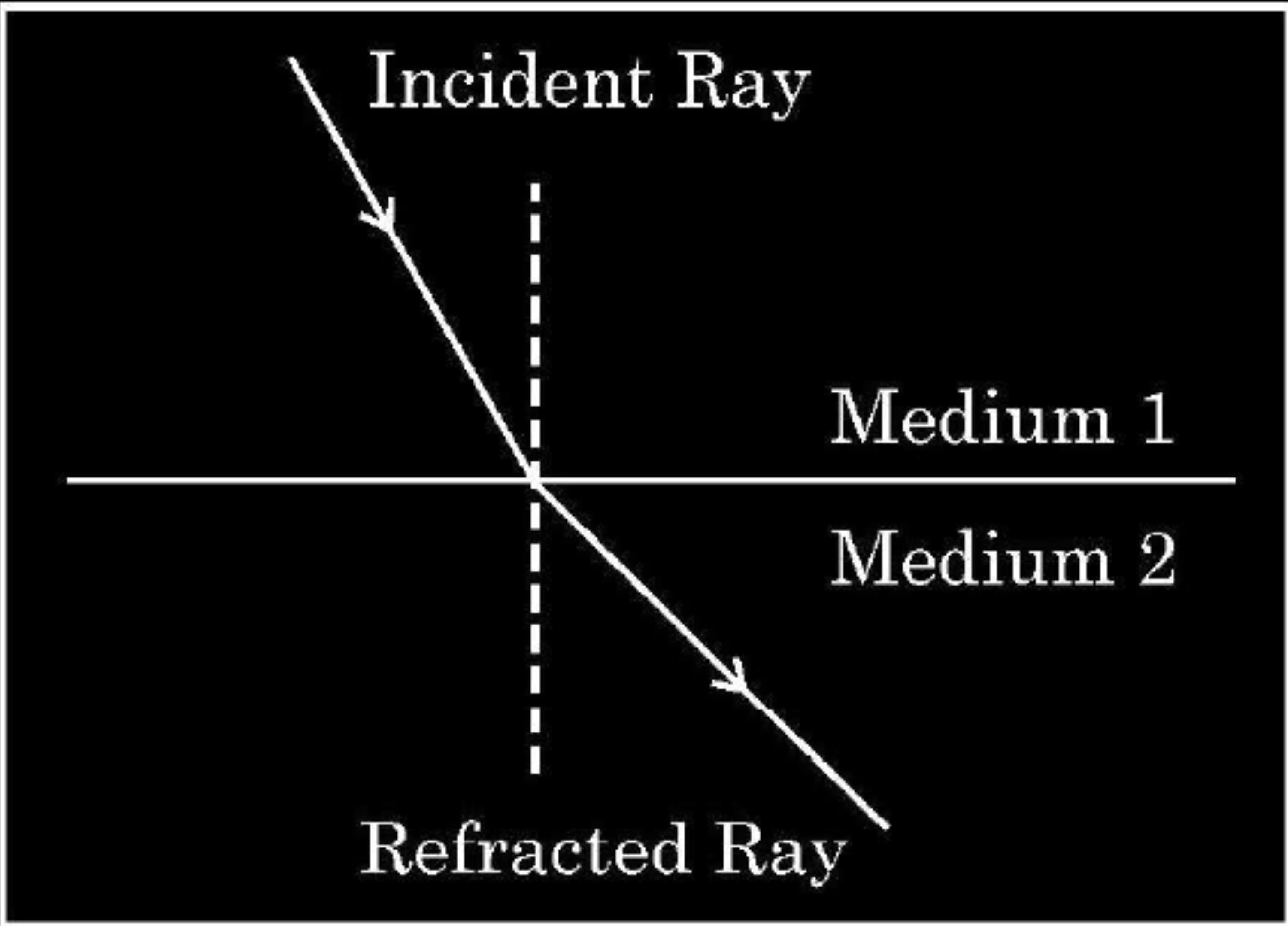
22. Study the figure in which the path of a ray of light going from Medium 1 to Medium 2 is shown.



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- (a) Out of the two Media – Medium 1 and Medium 2, in which is the speed of light more ?
- (b) State reason of bending of the refracted ray away from the normal.
- (c) Express refractive index of Medium 2 with respect to Medium 1 in terms of speed of light in two media.

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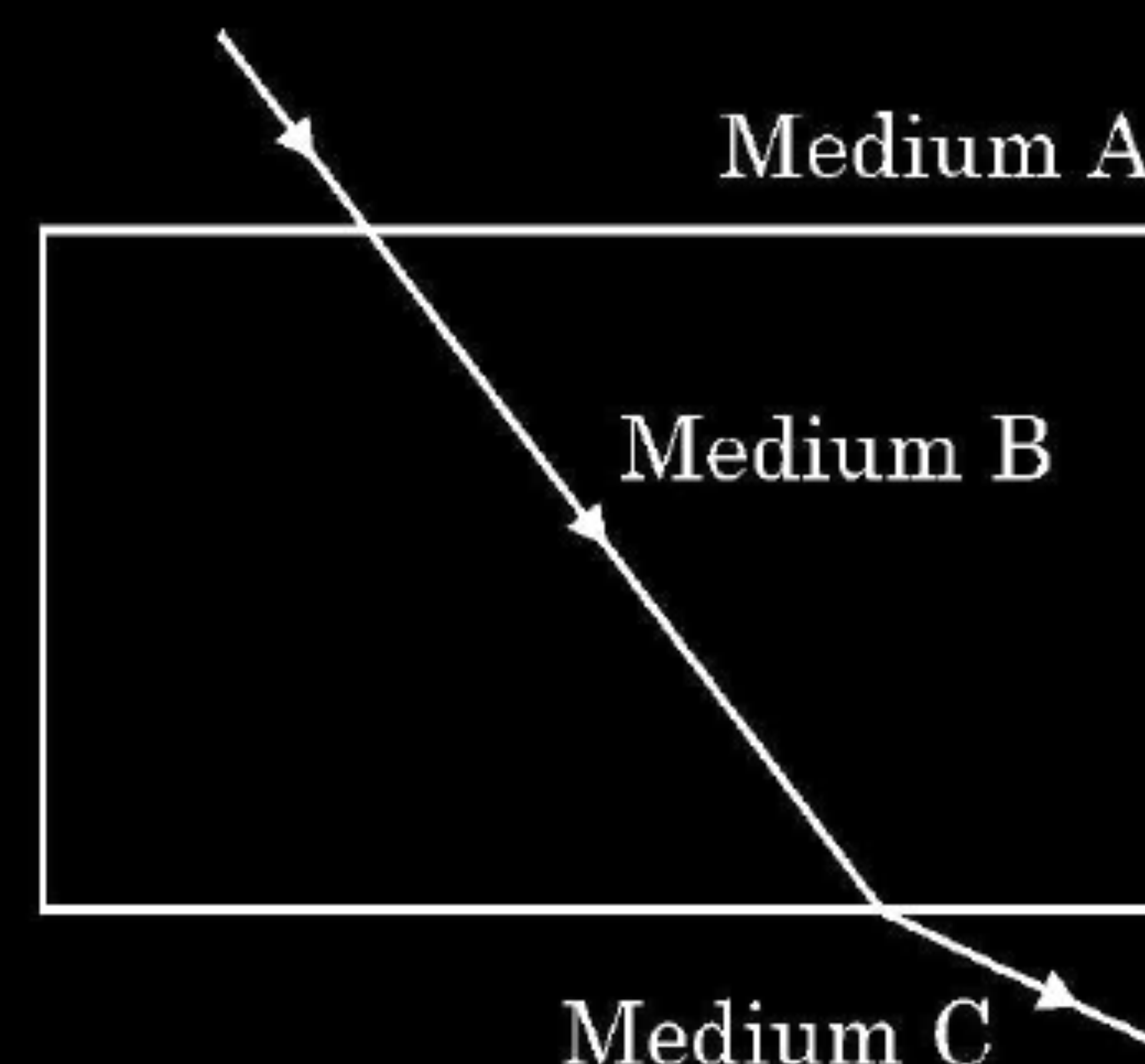
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- (a) Out of the two Media – Medium 1 and Medium 2, in which is the speed of light more ?
- (b) State reason of bending of the refracted ray away from the normal.
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2

(a) Medium 2	$\frac{1}{2}$
(b) Ray bends away from the normal because it is travelling from optically denser medium to rarer medium / speed of light in medium 2 increases.	$\frac{1}{2}$
(c) $n_{21} = \frac{\text{Speed of light in medium 1}}{\text{Speed of light in medium 2}}$	1

36. (A) (a) Observe the following diagram and compare (i) speed of light and (ii) optical densities of the three media A, B and C. Also give justification for your answer of any one of the two cases in terms of refractive indices of A, B and C.



- (b) Redraw the path of a ray of light through the three media, if the ray of light starting from medium A falls on the medium B
- (i) Obliquely and the optical density of medium B is made more than that of A and C.
 - (ii) The ray falls normally from medium A to medium B.

(A) (a) (i) Speed of light in A and B is same whereas the speed of light in C is greater than that of A and B.

$\frac{1}{2} + \frac{1}{2}$

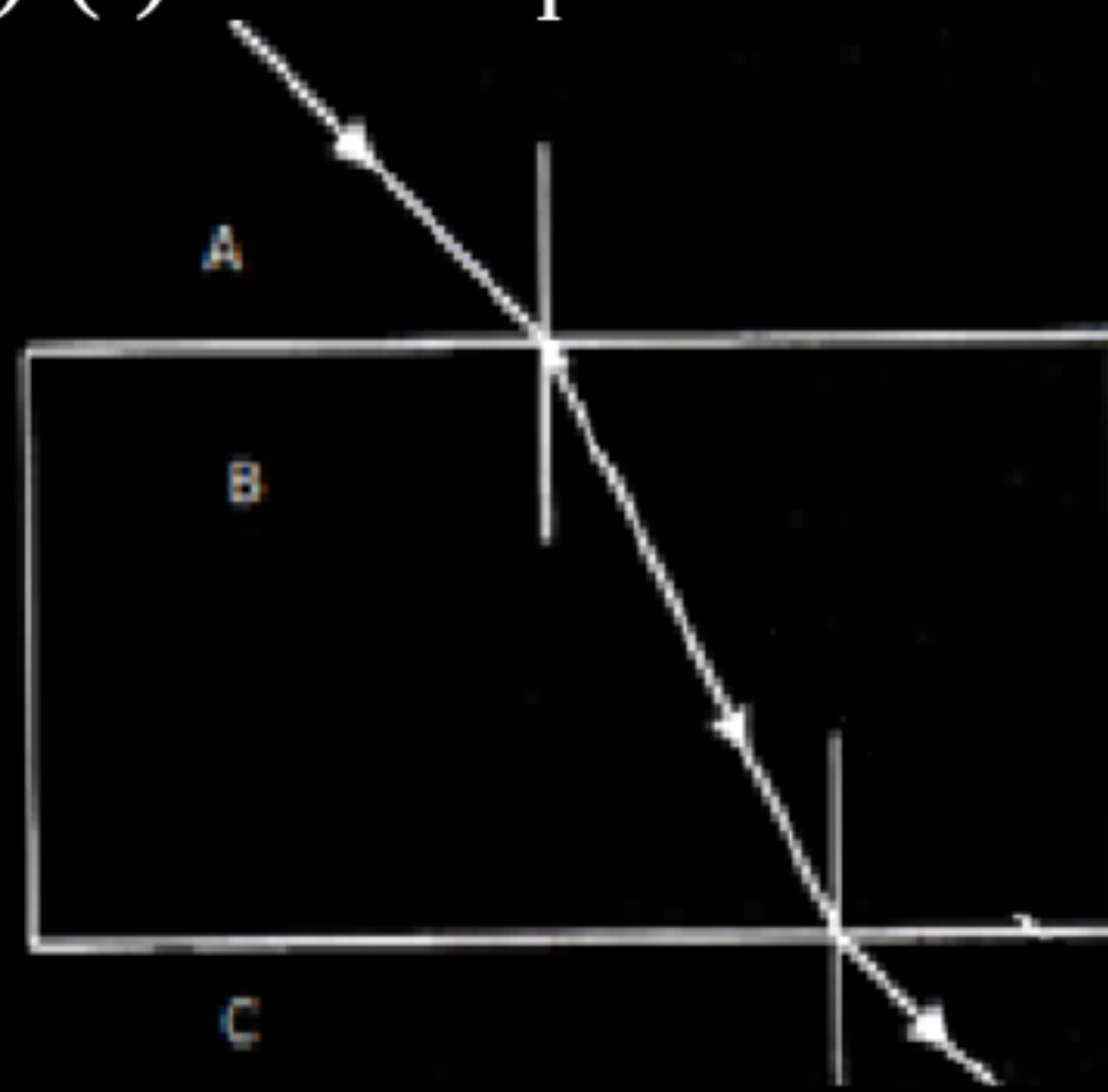
(ii) Optical density of A and B is same whereas optical density of C is less than that of A and B.

$\frac{1}{2} + \frac{1}{2}$

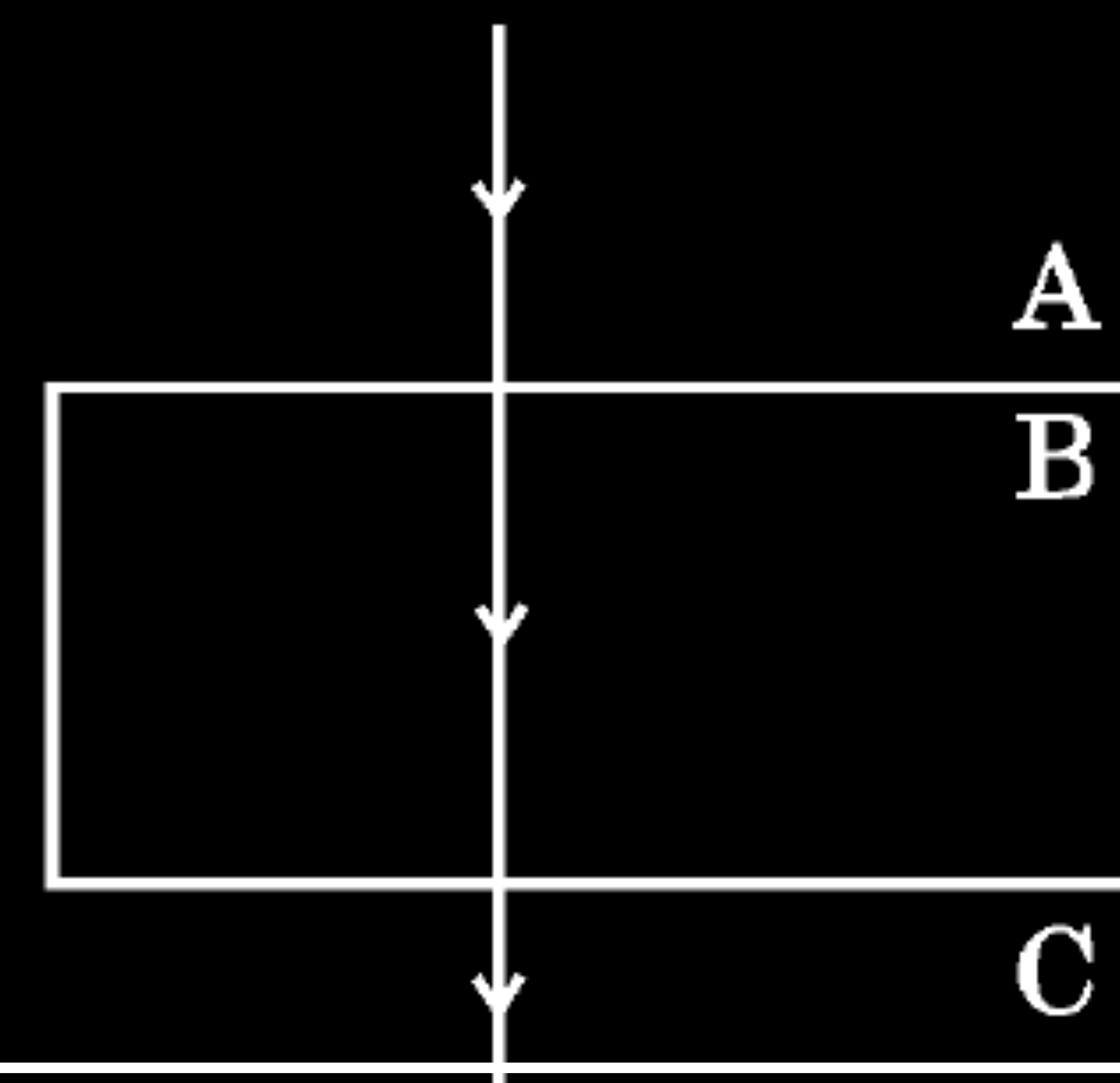
Since the ray of light does not bend while passing from A to B the refractive indices of A and B are same and since it bends away from the normal while passing from B to C the refractive index of C is less than that of A and B. / Refractive index of a medium is inversely proportional to the speed of light in that medium.

1

(b) (i) Oblique Incidence



(ii) Normal Incidence



1

1

23. When an incident ray of light enters a medium 'X' from air, it bends towards the normal.

Infer about the following with respect to air :

- (a) Optical density of medium 'X'.
- (b) Speed of light through medium 'X'

Give reason for your answer in each case.

23. When an incident ray of light enters a medium 'X' from air, it bends towards the normal.

Infer about the following with respect to air :

- (a) Optical density of medium 'X'.
- (b) Speed of light through medium 'X'

Give reason for your answer in each case.

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(a) Optical density of X is more than optical density of air because the ray coming from air bends towards the normal as it enters the medium X.	$\frac{1}{2} + \frac{1}{2}$
(b) Speed of light through medium X is less than the speed of light through air because X is optically denser than air.	$\frac{1}{2} + \frac{1}{2}$

24. The values of absolute refractive indices of kerosene and water are 1.44 and 1.33 respectively. Compare the two media on the basis of their

- (a) optical density
- (b) mass density
- (c) relative speed of propagation of light.

What do you infer on the basis of above comparisons ?

24. The values of absolute refractive indices of kerosene and water are 1.44 and 1.33 respectively. Compare the two media on the basis of their

- (a) optical density
- (b) mass density
- (c) relative speed of propagation of light.

What do you infer on the basis of above comparisons ?

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Kerosene (1.44)	Water (1.33)
(a) Higher optical density	Lower optical density
(b) Lower mass density	Higher mass density
(c) Lower speed of light	Higher speed of light

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- Inference: Although, kerosene is optically denser than water but its mass density is less than water/ An optically denser medium may not possess greater mass density.

$\frac{1}{2}$

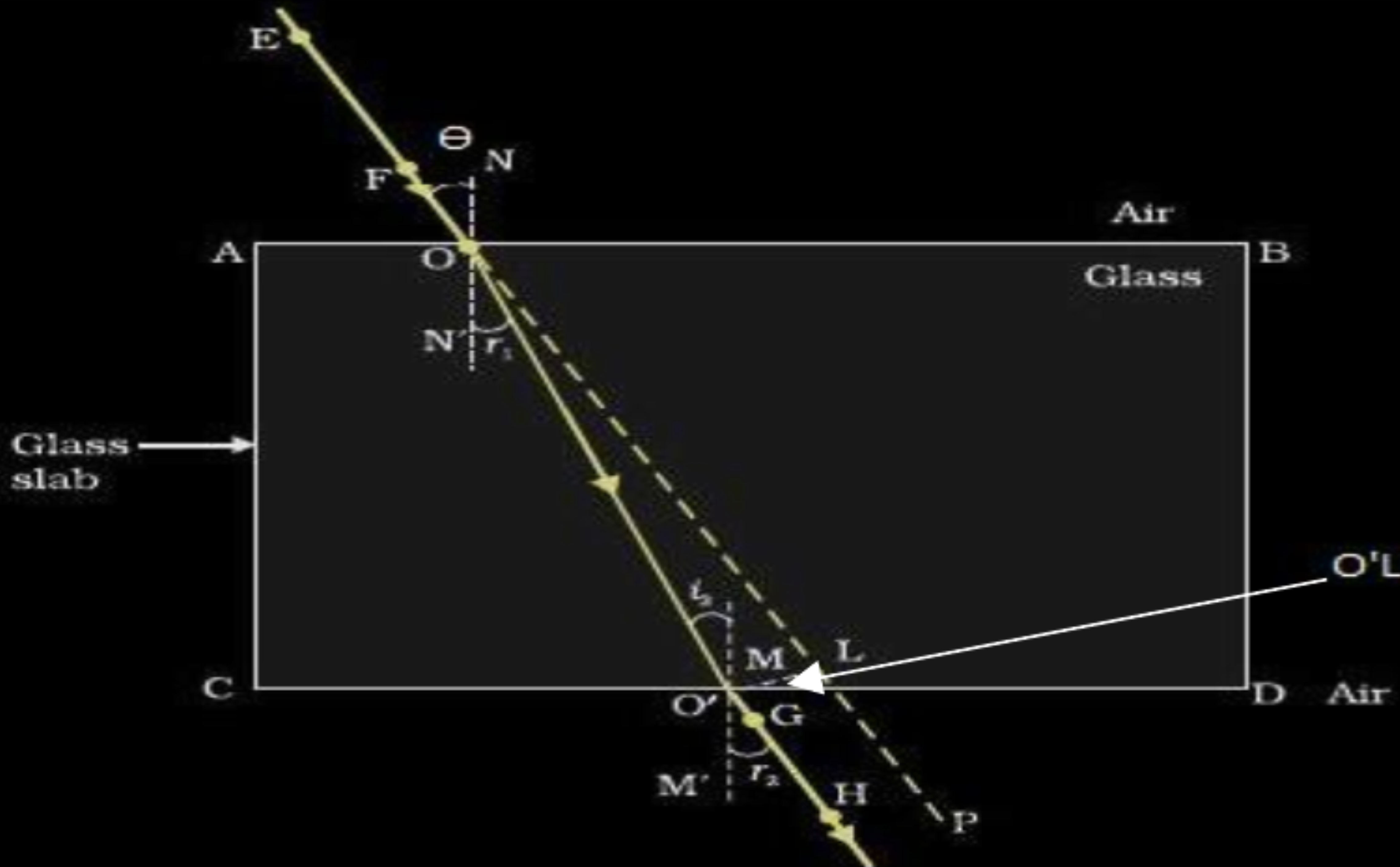
- 23.** A ray of light falls making an angle of incidence θ on the surface of a glass slab. Draw a labelled ray diagram to show its path. Also mark lateral displacement on it.

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2

 The diagram illustrates the refraction of light through a rectangular glass slab. An incident ray EF enters the top surface AB at point O. The normal at O is NN'. The angle of incidence is θ. The ray refracts into the glass, becoming ray OG. At the bottom surface CD, the ray emerges as ray GH. The normal at G is MM'. The angle of emergence is i_2. A dashed line EP represents the extension of the incident ray. The lateral displacement is the perpendicular distance OL between the incident ray's path and the emergent ray. Labels include 'Air' above and below the slab, 'Glass' inside, and 'Glass slab' with an arrow pointing to the slab. Points A, B, C, D mark the corners of the slab. Points E, F, O, G, H, M, L, P mark specific locations on the rays and normals. Angles i_1 and i_2 are also indicated.	1½
Lateral displacement labelling	½

15. The speed of light in vacuum is 3×10^8 m/s. If the speed of light in a medium is 2.25×10^8 m/s, the absolute refractive index of the medium is : **1**

(A) $\frac{7}{6}$

(B) $\frac{5}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

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15. The speed of light in vacuum is 3×10^8 m/s. If the speed of light in a medium is 2.25×10^8 m/s, the absolute refractive index of the medium is : **1**

(A) $\frac{7}{6}$

(B) $\frac{5}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

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(C)/ 4/3

24. (a) State two laws of refraction of light.

2

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(a) Laws of Refraction of light :

(i) The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.

1

(ii) The ratio of the sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for the given pair of media.

1

Note:

If a student writes $\frac{\sin i}{\sin r} = \text{constant}$ instead of statement, award ½ mark only)

- (b) Define the term absolute refractive index of a medium. A ray of light enters from vacuum to glass of absolute refractive index 1.5. Find the speed of light in glass. The speed of light in vacuum is 3×10^8 m/s. **2**

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- (b) Define the term absolute refractive index of a medium. A ray of light enters from vacuum to glass of absolute refractive index 1.5. Find the speed of light in glass. The speed of light in vacuum is 3×10^8 m/s.

2

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(b) Absolute refractive index of a medium is the ratio of speed of light in vacuum to the speed of light in the medium.

Given:

$$c = 3 \times 10^8 \text{ m/s}; n_m = 1.5; v_m = ?$$

Absolute refractive index of a medium (n_m)

$$= \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} = \frac{c}{v_m}$$

$$v_m = \frac{c}{n_m} = 2 \times 10^8 \text{ m/s}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

11. Absolute refractive index of glass and water is $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4} \times 10^8$ m/s

(B) $\frac{5}{2} \times 10^8$ m/s

(C) $\frac{7}{3} \times 10^8$ m/s

(D) $\frac{16}{9} \times 10^8$ m/s

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(D) $\frac{16}{9} \times 10^8$ m/s

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(A)/9/4 x 10⁸ m/s

(b) (i) State laws of refraction of light.

(ii) Draw a ray diagram to show refraction of a ray of light through a rectangular glass slab. How is the emergent ray related to incident ray ? Mark lateral displacement in the diagram.

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(ii) Draw a ray diagram to show refraction of a ray of light through a rectangular glass slab. How is the emergent ray related to incident ray ? Mark lateral displacement in the diagram.

5

(b)(i) -The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane. - The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for the given pair of media. / $\frac{\sin i}{\sin r} = \text{constant}$	1 1
(ii) • The emergent ray is parallel to the incident ray. • Labelling of lateral displacement (If labelling is not done deduct ½ marks)	2 ½ ½

39. When a ray of light moving in a medium enters obliquely into another medium, it bends from its path. This phenomenon is called refraction of light. The ability of a medium to refract light is also expressed in terms of optical density. It is not the same as mass density. We use the terms ‘rarer medium’ and ‘denser medium’ which actually means ‘optically rarer medium’ and ‘optically denser medium’ respectively. When we say that a medium ‘A’ is optically denser than the other medium ‘B’, we mean that the refractive index of medium A is more than the refractive index of medium ‘B’. The speed of light is higher in a rarer medium than a denser medium. Thus a ray of light travelling from a rarer medium to a denser medium slows down and bends towards the normal.

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- (a) Define the term absolute refractive index of a medium. 1
- (b) Absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively.
- (i) In which one of the two media is the speed of light more ?
- (ii) If a ray of light enters obliquely from glass to water, will it bend towards the normal or away from the normal ? 1

- (c) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find the speed of light in (i) water and (ii) vacuum. 2

OR

- (c) “A ray of light incident on a rectangular glass slab immersed in any medium emerges parallel to itself.” Draw a labelled ray diagram to justify this statement. CBSE 2023 C 2

(a) The refractive index of a medium with respect to air or vacuum
 / Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$

1

(b) (i) speed of light is more in water
 (ii) bends away from normal

$\frac{1}{2}$

$\frac{1}{2}$

(c) Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$

$$\frac{3}{2} = \frac{\text{speed of light in vacuum}}{2 \times 10^8}$$

$\frac{1}{2}$

Speed of light in vacuum = 3×10^8 m/s

$\frac{1}{2}$

Speed of light in water = $\frac{3}{4} \times 3 \times 10^8$ m/s = $\frac{9}{4} \times 10^8$ m/s

1

OR

(c)

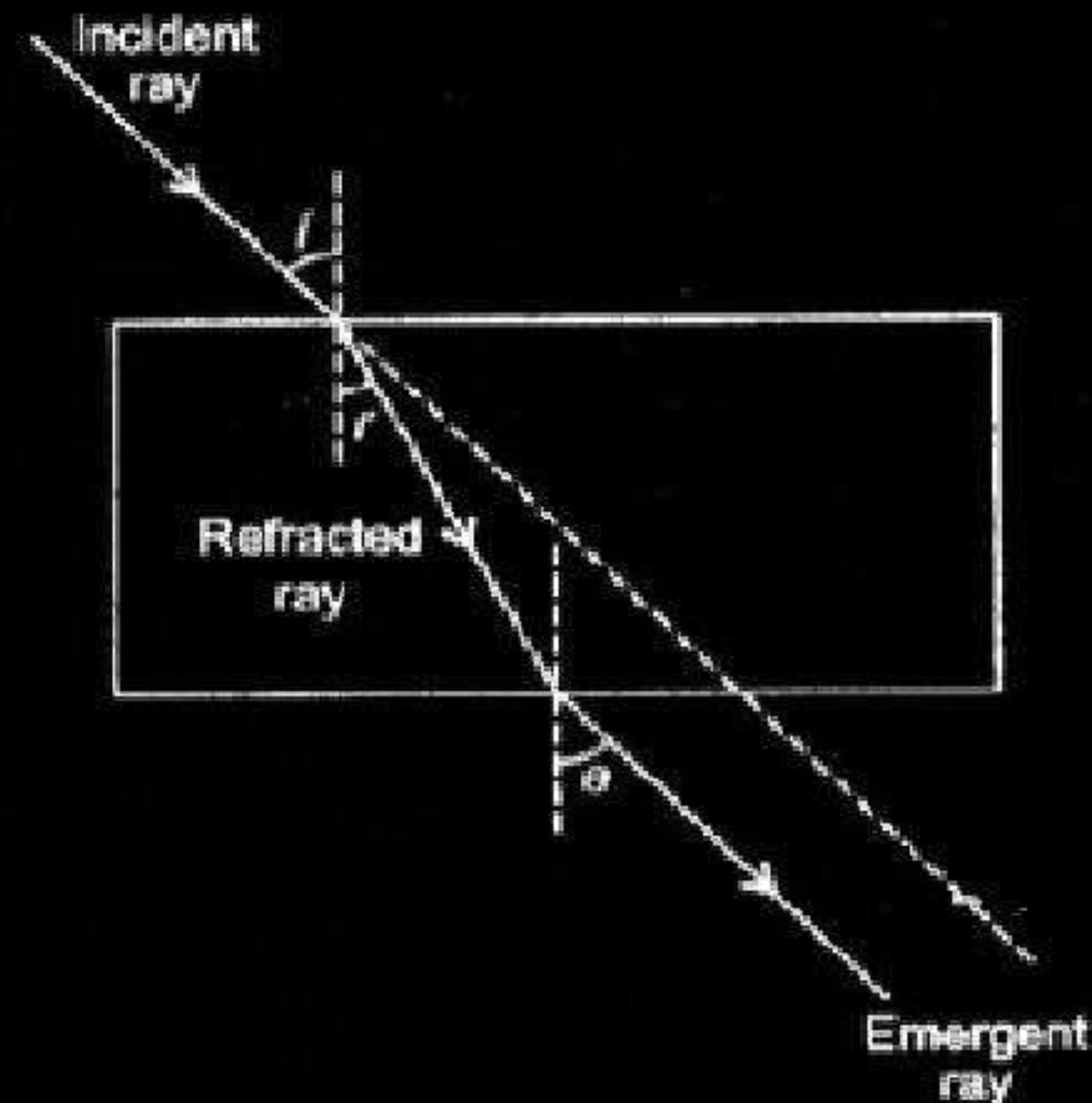


Diagram
 Labelling

1

1

37. When a ray of light moving in a medium enters obliquely in another medium, it bends from its path. This phenomenon is called refraction of light. The ability of a medium to refract light is expressed in terms of optical density. The term optical density is not the same as mass density. If we compare two optical media, the one with the larger absolute refractive index is an optically denser medium than the other. Also, the speed of light through a given medium is inversely proportional to its optical density.

- (a) Define the term 'absolute refractive' index of a medium. 1
- (b) Refractive indices of glass, water and carbon disulphide are 1.50, 1.33 and 1.62 respectively. In which one of these media is the speed of light (i) highest, and (ii) lowest ? 1
- (c) (i) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in water is 2.25×10^8 m/s, find the speed of light in (I) vacuum, and (II) glass. 2

OR

- (ii) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. Find the refractive index of water with respect to glass. 2

(a) The refractive index of a medium with respect to air or vacuum /	
Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$	1
(b) (i) water (ii) carbon disulphide	$\frac{1}{2} + \frac{1}{2}$
(c) (i)	
$n_g = \frac{3}{2}; n_w = \frac{4}{3}; v_w = 2.25 \times 10^8 \text{ m/s}; c = ?; v_g = ?$	
(I) $n_w = \frac{c}{v_w}$	$\frac{1}{2}$
$\frac{4}{3} = \frac{c}{2.25 \times 10^8 \text{ m/s}}$	
$c = 3 \times 10^8 \text{ m/s}$	$\frac{1}{2}$
(II) $v_g = \frac{c}{n_g}$	$\frac{1}{2}$
$= \frac{3 \times 10^8 \text{ m/s}}{3/2} = 2 \times 10^8 \text{ m/s}$	$\frac{1}{2}$
OR	
(ii)	
Refractive index of water w.r.t. glass = $\frac{\text{speed of light in glass}}{\text{speed of light in water}}$	
Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}}$	$\frac{1}{2}$
$\therefore \text{speed of light in glass} = \frac{c}{\text{Absolute refractive index of glass}} = \frac{c}{3/2} = \frac{2c}{3}$	$\frac{1}{2}$
$\text{speed of light in water} = \frac{c}{\text{Absolute refractive index of water}} = \frac{c}{4/3} = \frac{3c}{4}$	$\frac{1}{2}$
$\therefore \text{Refractive index of water w. r. t. glass} = \frac{2c/3}{3c/4} = \frac{8}{9}$	$\frac{1}{2}$

- (b) (i) Define absolute refractive index.
- (ii) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find :
- (I) Speed of light in vacuum
- (II) Speed of light in water

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(b) (i) Define absolute refractive index.

(ii) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find :

(I) Speed of light in vacuum

(II) Speed of light in water

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3

(b)

(i) Absolute refractive index of a medium is the ratio of speed of light in vacuum (or air) to the speed of light in the medium.

$$\text{Absolute refractive index } n_m = \frac{\text{Speed of light in vacuum } (c)}{\text{Speed of light in the medium } v_m}$$

(ii) (I) $c = n_g \times v_g$

$$= \frac{3}{2} \times 2 \times 10^8 \text{ m/s} = 3 \times 10^8 \text{ m/s}$$

(II) $v_w = \frac{c}{n_w}$

$$= \frac{3 \times 10^8 \text{ m/s}}{\frac{4}{3}} = \frac{9}{4} \times 10^8 \text{ m/s or } 2.25 \times 10^8 \text{ m/s}$$

1

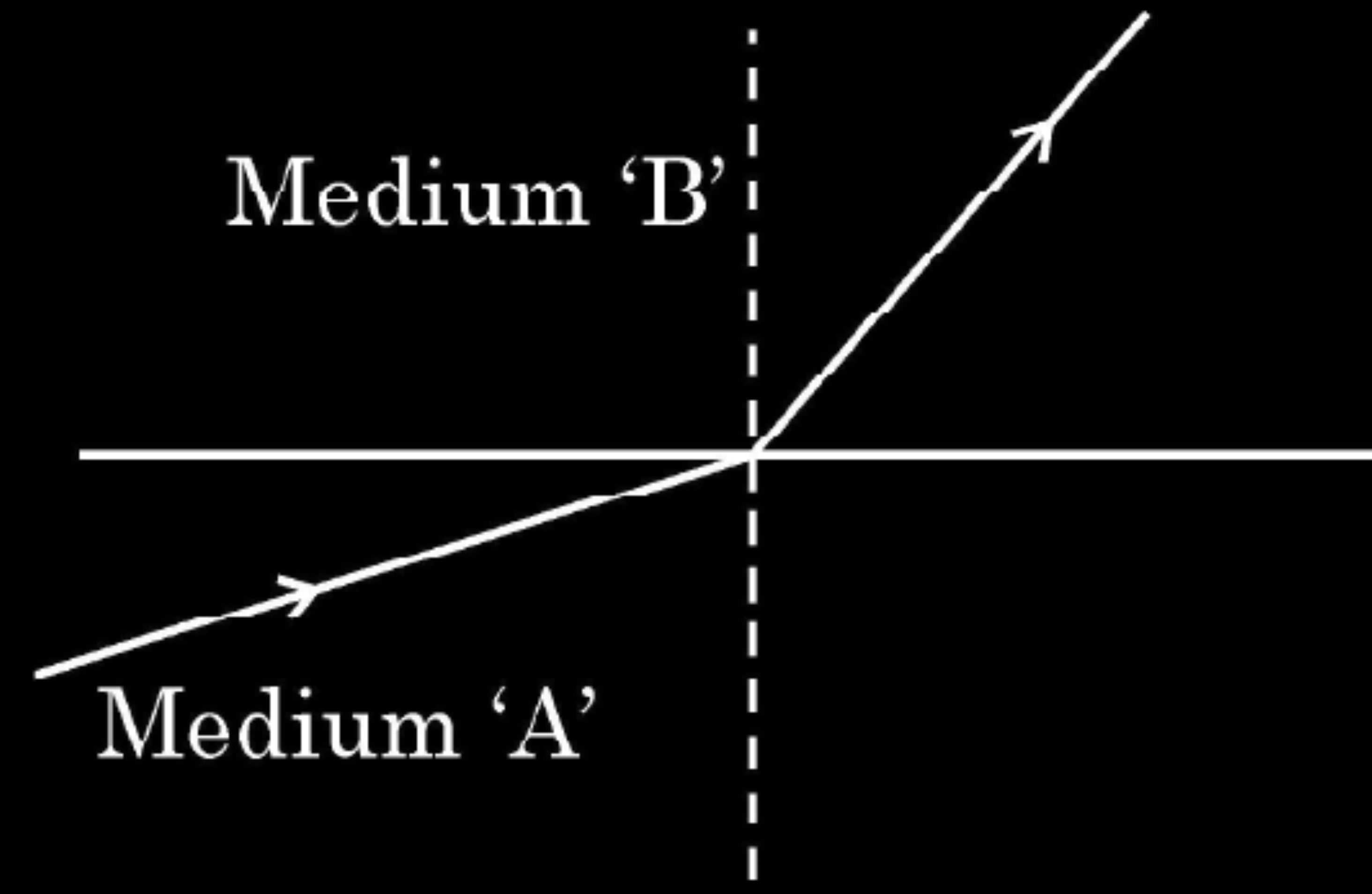
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

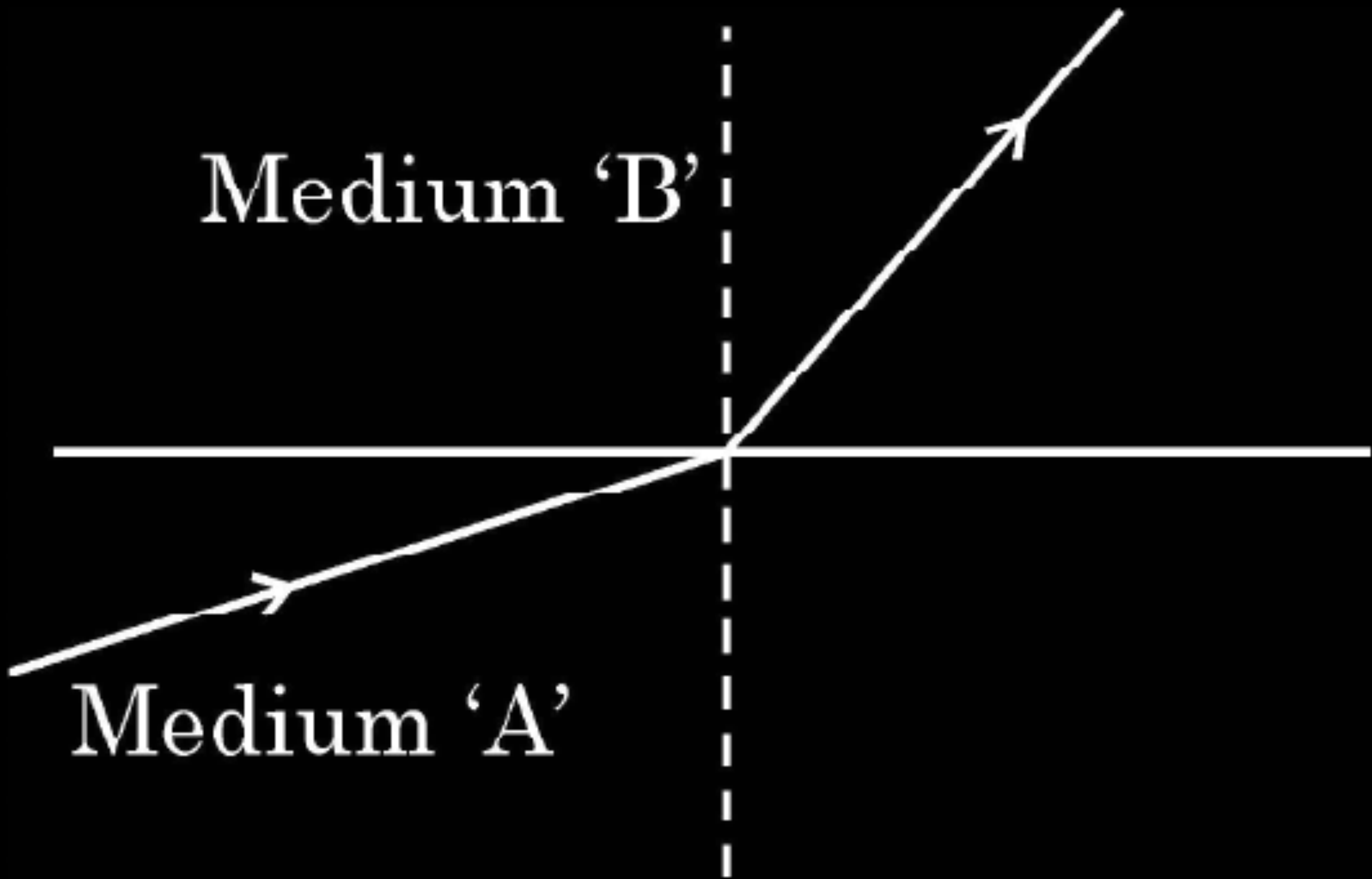
$\frac{1}{2}$

25. A light ray enters from medium A to medium B as shown in the figure.



- (a) Which one of the two media is denser w.r.t. other medium ? Justify your answer. 1
- (b) If the speed of light in medium A is v_a and in medium B is v_b , what is the refractive index of B with respect to A. 1

25. A light ray enters from medium A to medium B as shown in the figure.



- (a) Which one of the two media is denser w.r.t. other medium ? Justify your answer.

1
- (b) If the speed of light in medium A is v_a and in medium B is v_b , what is the refractive index of B with respect to A.

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1

(a) Medium B	$\frac{1}{2}$
In medium B ray of light bends towards normal / $\angle r < \angle i$	$\frac{1}{2}$
(b) Refractive index of Medium 'B' with respect of Medium 'A' is	
$n_{BA} = \frac{v_a}{v_b}$	1

- (a) A ray of light starting from diamond is incident on the interface separating diamond and water. Draw a labelled ray diagram to show the refraction of light in this case. 1
- (b) Absolute refractive indices of diamond and water are 2.42 and 1.33 respectively. Find the value of refractive index of water w.r.t. diamond. 1

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(a) A ray of light starting from diamond is incident on the interface separating diamond and water. Draw a labelled ray diagram to show the refraction of light in this case.

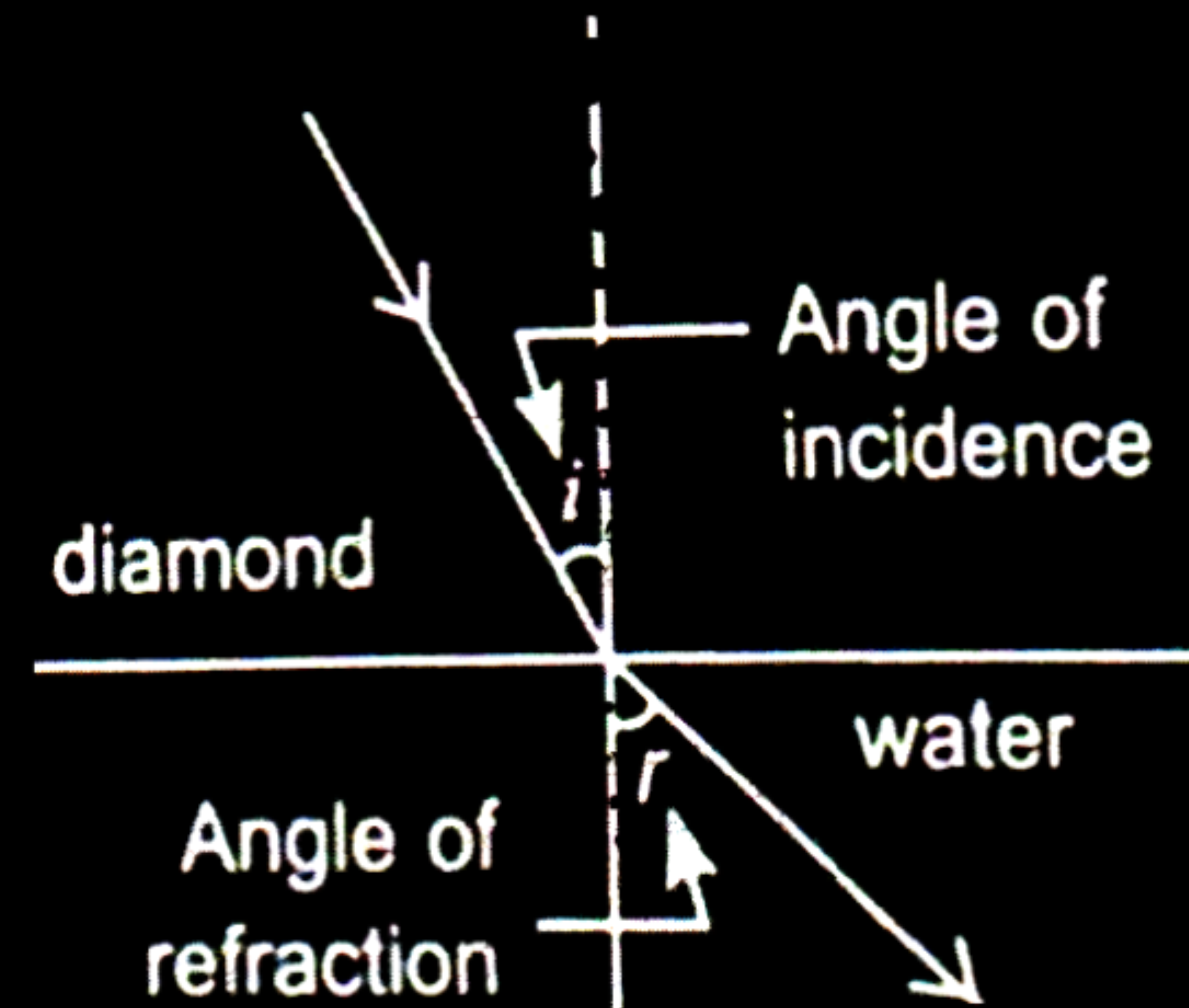
1

(b) Absolute refractive indices of diamond and water are 2.42 and 1.33 respectively. Find the value of refractive index of water w.r.t. diamond.

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1

(a)



(Credit marks for $\angle i$, $\angle r$ and arrows.)

1

(b)

$$n_{21} = \frac{n_{2a}}{n_{1a}}$$

$$\frac{1.33}{2.42} \quad \text{or} \quad 0.55$$

$\frac{1}{2}$

$\frac{1}{2}$

39. The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the large refractive index is optically denser medium than the other. The other medium with a lower refractive index is optically rarer. Also the speed of light through a given medium is inversely proportional to its optical density. 4
- (i) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is 3×10^8 m/s. 1
- (ii) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say θ), then write the increasing order of the angle of refraction in these media. 1
- (iii) (A) The speed of light in glass is 2×10^8 m/s and in water is 2.25×10^8 m/s. 2
- (a) Which one of the two is optically denser and why ?
- (b) A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass ? Give reason.

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OR

- (iii) (B) The absolute refractive indices of water and glass are $4/3$ and $3/2$ respectively. If the speed of light in glass is 2×10^8 m/s, find the speed of light in (i) vacuum and (ii) water. 2

(i) Refractive index of diamond = $\frac{\text{Speed of light in vacuum}}{\text{Speed of light in diamond}}$	$\frac{1}{2}$
Speed of light in diamond = $\frac{3 \times 10^8 \text{ m/s}}{2.42} = 1.23 \times 10^8 \text{ m/s}$	$\frac{1}{2}$
(ii) $\angle r$ in carbon disulphide < $\angle r$ in glass < $\angle r$ in water	1
(iii) (A)	$\frac{1}{2}$
(a) • Glass	
• The speed of light in water is more than the speed of light in glass. / Refractive index of glass is more than the refractive index of water.	$\frac{1}{2}$
(b) Light will enter from water to glass without bending (undeviated / straight) because in this case $\angle i = 0$; $\angle r = 0$.	1
OR	
(iii) (B)	
$n_{\text{glass}} = \frac{3}{2}$	
$n_{\text{water}} = \frac{4}{3}$	
$v_{\text{glass}} = 2 \times 10^8 \text{ m/s}$	
$n_{\text{glass}} = \frac{\text{speed of light in vacuum}(c)}{\text{speed of light in glass}(v_g)}$	
$c = n_{\text{glass}} \times v_{\text{glass}}$ $= \frac{3}{2} \times 2 \times 10^8 \text{ m/s}$	
$= 3 \times 10^8 \text{ m/s}$	1
$v_{\text{water}} = \frac{c}{n_{\text{water}}} = \frac{3 \times 10^8 \text{ m/s}}{\frac{4}{3}}$	
$= \frac{9}{4} \times 10^8 \text{ m/s}$ or $2.25 \times 10^8 \text{ m/s}$	1

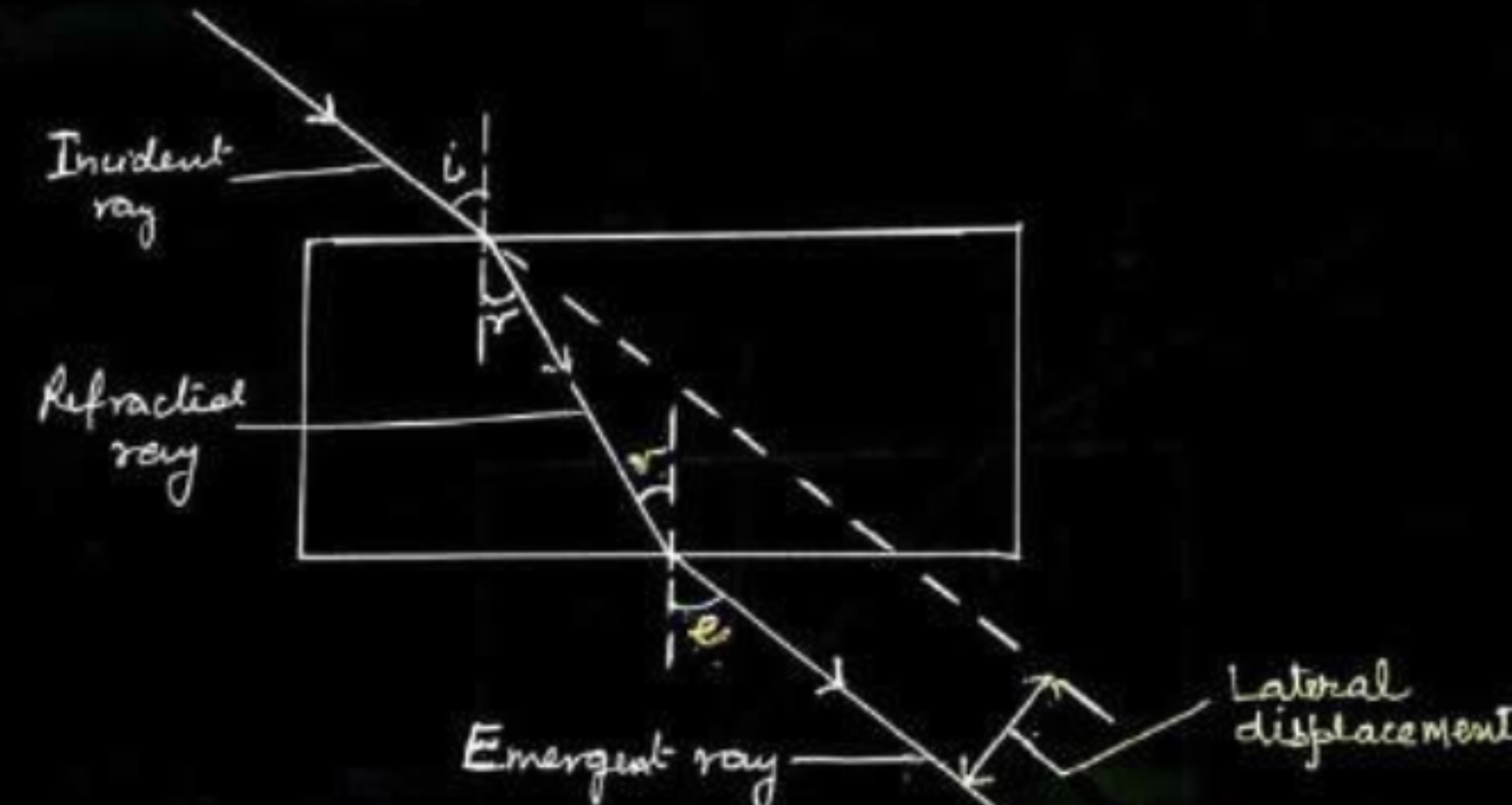
The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the large refractive index is optically denser medium than the other. The other medium with a lower refractive index is optically rarer. Also the speed of light through a given medium is inversely proportional to its optical density.	4
(i) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is $3 \times 10^8 \text{ m/s}$.	1
(ii) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say θ), then write the increasing order of the angle of refraction in these media.	1
(iii) (A) The speed of light in glass is $2 \times 10^8 \text{ m/s}$ and in water is $2.25 \times 10^8 \text{ m/s}$.	2
(a) Which one of the two is optically denser and why ?	
(b) A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass ? Give reason.	
OR	
(iii) (B) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is $2 \times 10^8 \text{ m/s}$, find the speed of light in (i) vacuum and (ii) water.	2

- 30.** (a) Draw a labelled ray diagram to show the path of a ray of light incident obliquely on one face of a glass slab.
- (b) Calculate the refractive index of the material of a glass slab. Given that the speed of light through the glass slab is 2×10^8 m/s and in air is 3×10^8 m/s.
- (c) Calculate the focal length of a lens, if its power is -2.5 D. CBSE 2020 5

- 30.** (a) Draw a labelled ray diagram to show the path of a ray of light incident obliquely on one face of a glass slab.
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- (c) Calculate the focal length of a lens, if its power is -2.5 D. **CBSE 2020**

5

(a) .



(Note –Deduct $\frac{1}{2}$ mark if arrows are not shown)

$$(b) n_{ga} = \frac{\text{Speed of light in air}}{\text{Speed of light in glass}} = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2} = 1.5$$

$$(c) f (m) = \frac{1}{P (D)}$$

$$f = \frac{1}{P} = \frac{1}{-2.5D} = \frac{-10}{25D} = -0.4m$$

(Note –Deduct $\frac{1}{2}$ marks if unit is not mentioned)

2

$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

- 22.** The refractive index of a medium 'x' with respect to a medium 'y' is $\frac{2}{3}$ and the refractive index of medium 'y' with respect to medium 'z' is $\frac{4}{3}$. Find the refractive index of medium 'z' with respect to medium 'x'. If the speed of light in medium 'x' is $3 \times 10^8 \text{ ms}^{-1}$, calculate the speed of light in medium 'y'.

CBSE 2020

22. The refractive index of a medium 'x' with respect to a medium 'y' is $\frac{2}{3}$ and the refractive index of medium 'y' with respect to medium 'z' is $\frac{4}{3}$. Find the refractive index of medium 'z' with respect to medium 'x'. If the speed of light in medium 'x' is $3 \times 10^8 \text{ ms}^{-1}$, calculate the speed of light in medium 'y'.

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3

$$\bullet \quad n_{xy} = \frac{2}{3} \quad \therefore \quad n_{yx} = \frac{3}{2}$$

$$n_{yz} = \frac{4}{3} \quad \therefore \quad n_{zy} = \frac{3}{4}$$

$$n_{zx} = n_{zy} \times n_{yx}$$

$$\therefore \quad n_{zx} = \frac{3}{4} \times \frac{3}{2} = \frac{9}{8}$$

$$\bullet \quad n_{yx} = \frac{v_x}{v_y}$$

$$\frac{3}{2} = \frac{3 \times 10^8}{v_y}$$

$$v_y = \frac{3 \times 10^8 \times 2}{3} = 2 \times 10^8 \text{ m/s}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

The absolute refractive index of Ruby is 1.7. Find the speed of light in Ruby. The speed of light in vacuum is 3×10^8 m/s.

CBSE 2019

The absolute refractive index of Ruby is 1.7. Find the speed of light in Ruby. The speed of light in vacuum is 3×10^8 m/s.

CBSE 2019

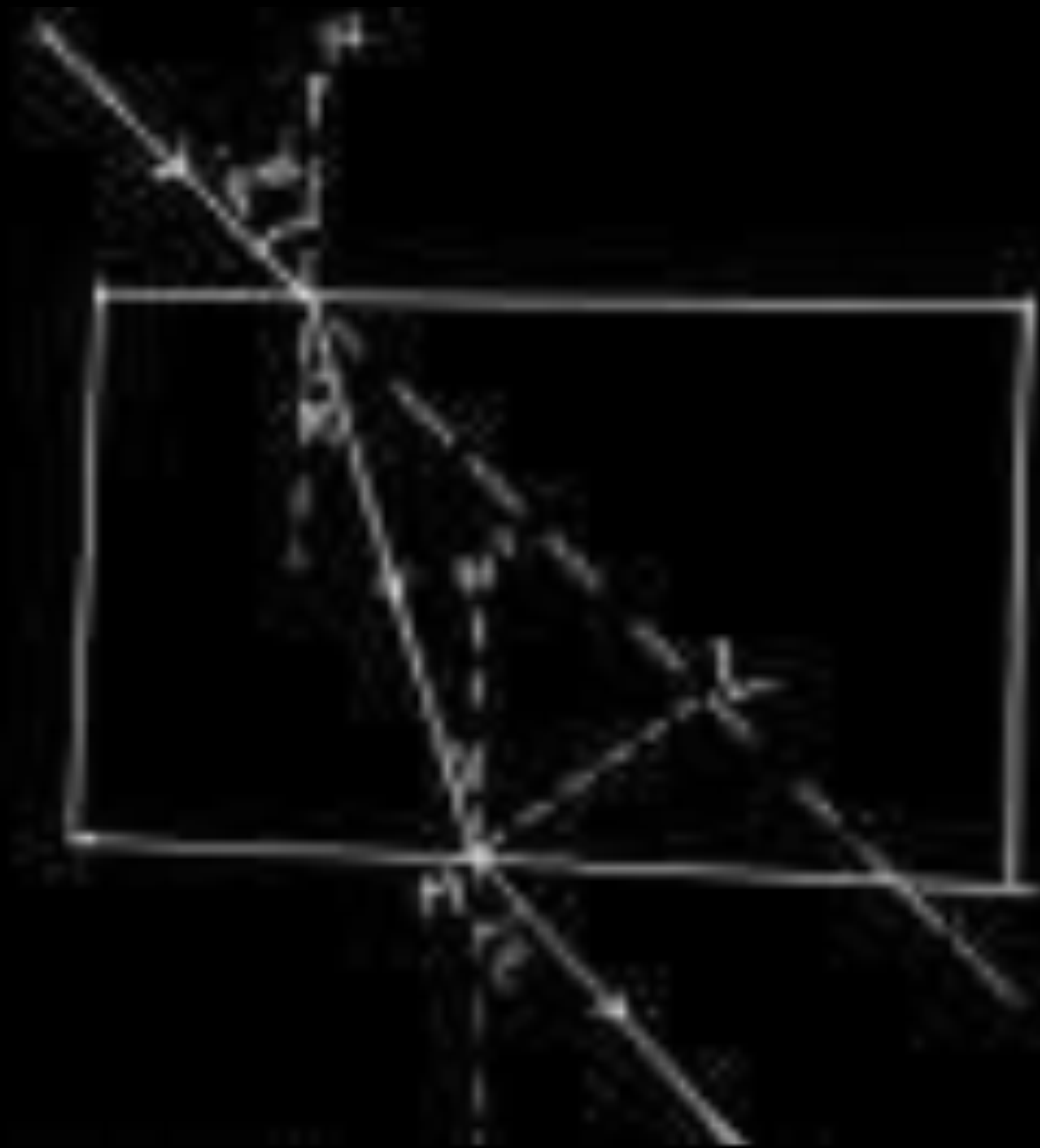
$\mu = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in Ruby}} = \frac{c}{v}$	$\frac{1}{2}$
$V = \frac{c}{\mu}$ c= velocity of light μ = refractive index	$\frac{1}{2}$
$v = \frac{3 \times 10^8}{1.7} = 1.76 \times 10^8 \text{ m/s}$	$\frac{1}{2} + \frac{1}{2}$

Draw the path of a ray of light when it enters one of the faces of a glass slab at an angle of nearly 45° . Label on it (i) angle of refraction, (ii) angle of emergence and (iii) lateral displacement.

CBSE 2019

Draw the path of a ray of light when it enters one of the faces of a glass slab at an angle of nearly 45° . Label on it (i) angle of refraction, (ii) angle of emergence and (iii) lateral displacement.

CBSE 2019



Labelling

- Angle of refraction (r_1)
- Angle of emergence (e)
- Lateral displacement (ML)

$\frac{1}{2}$

$\frac{1}{2} \times 3$

A student has to trace the path of a ray of light passing through a rectangular glass slab for four different values of angle of incidence.

- (a) Write two important precautions for this experiment.
- (b) List two conclusions the student will draw based on his experiment.

A student has to trace the path of a ray of light passing through a rectangular glass slab for four different values of angle of incidence.

- (a) Write two important precautions for this experiment.
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CBSE 2019

The student should take the following precaution

(a) Precaution -

- (i) See that the pins are in a straight line and atleast 3cm apart.
- (ii) Angle of incidence should be between 30^0 to 60^0 .
- (iii) Glass slab should always remain inside the boundary.

(any two)

$\frac{1}{2}$

$\frac{1}{2}$

(b) Conclusion -

- (i) The emergent ray is parallel to incident ray
- (ii) Lateral displacement takes place.
- (iii) Angle of incidence = Angle of emergence

(any two)

$\frac{1}{2}$

$\frac{1}{2}$

State laws of reflection of light.

CBSE 2019

OR

Define absolute refractive index and express it mathematically.

State laws of reflection of light.

CBSE 2019

OR

Define absolute refractive index and express it mathematically.

- **First law:** The incident ray, the reflected ray and the normal to the surface at the point of incidence all lie in the same plane.
 - **Second law:** Angle of incidence is equal to angle of reflection.
- OR
- Refractive index of a transparent medium with respect to vacuum or air is called absolute refractive index.
 - Mathematically

$$\text{Absolute refractive index of a medium} = \frac{\text{Speed of light in vacuum or air}}{\text{Speed of light in the medium}}$$
$$n = c/v$$

1

1

1

1

State laws of refraction of light.

CBSE 2019

OR

List four characteristics of the image formed by a concave mirror of focal length 40 cm when the object is placed in front of it at a distance of 20 cm from its pole.

State laws of refraction of light.

CBSE 2019

OR

List four characteristics of the image formed by a concave mirror of focal length 40 cm when the object is placed in front of it at a distance of 20 cm from its pole.

First law :The incident ray ,the refracted ray and the normal to the interface of two mediums at the point of incidence, all lie in the same plane.

1

Second law :The ratio of sine of angle of incidence to sine of angle of refraction is always constant for a given pair of medium .This constant is called refractive index.

1

OR

(i)Erect (ii) virtual (iii) magnified (iv) Behind the mirror

$\frac{1}{2} \times 4$

Define refractive index of a transparent medium. The speed of light in a medium of absolute refractive index 1.5 is $2 \times 10^8 \text{ ms}^{-1}$. What is the speed of light in vacuum ?

CBSE 2019

Define refractive index of a transparent medium. The speed of light in a medium of absolute refractive index 1.5 is $2 \times 10^8 \text{ ms}^{-1}$. What is the speed of light in vacuum ?

CBSE 2019

- Refractive index is the ratio of speed of light in air or vacuum to speed of light in medium.

Mathematically: Refractive index of a medium 1.5

Speed of light in medium = $2 \times 10^8 \text{ ms}^{-1}$

Speed of light in vacuum?

$\mu = \text{speed of light in vacuum} / \text{speed of light in medium}$

speed of light in vacuum = $1.5 \times 2 \times 10^8 \text{ ms}^{-1}$

$3 \times 10^8 \text{ ms}^{-1}$

1

1

State the laws of refraction of light. Explain the term ‘absolute refractive index of a medium’ and write an expression to relate it with the speed of light in vacuum.

CBSE 2018

State the laws of refraction of light. Explain the term ‘absolute refractive index of a medium’ and write an expression to relate it with the speed of light in vacuum.

CBSE 2018	
<u>1st law</u> : the incident ray, refracted ray and normal to the interface at the point of incidence lie in the same plane.	1
<u>2nd law</u> : The sine of angle of incidence bears a constant ratio with sine of angle of refraction for a given pair of media. Or $\frac{\sin i}{\sin r} = \text{constant}$	1
Absolute Refractive Index of a medium = $\frac{\text{Speed of light in air or vacuum}}{\text{Speed of light in the medium}}$ (Award full marks if the same thing is given in the form of statement)	1

A student very cautiously traces the path of a ray through a glass slab for different values of the angle of incidence ($\angle i$). He then measures the corresponding values of the angle of refraction ($\angle r$) and the angle of emergence ($\angle e$) for every value of the angle of incidence. On analysing these measurements of angles, his conclusion would be

- (A) $\angle i > \angle r > \angle e$
- (B) $\angle i = \angle e > \angle r$
- (C) $\angle i < \angle r < \angle e$
- (D) $\angle i = \angle e < \angle r$

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CBSE 2017

In your laboratory you trace the path of light rays through a glass slab for different values of angle of incidence ($\angle i$) and in each case measure the values of the corresponding angle of refraction ($\angle r$) and angle of emergence ($\angle e$).

On the basis of your observations your correct conclusion is :

- (a) $\angle i$ is more than $\angle r$, but nearly equal to $\angle e$
- (b) $\angle i$ is less than $\angle r$, but nearly equal to $\angle e$
- (c) $\angle i$ is more than $\angle e$, but nearly equal to $\angle r$
- (d) $\angle i$ is less than $\angle e$, but nearly equal to $\angle r$

CBSE 2016

In your laboratory you trace the path of light rays through a glass slab for different values of angle of incidence ($\angle i$) and in each case measure the values of the corresponding angle of refraction ($\angle r$) and angle of emergence ($\angle e$).

On the basis of your observations your correct conclusion is :

- (a) $\angle i$ is more than $\angle r$, but nearly equal to $\angle e$
- (b) $\angle i$ is less than $\angle r$, but nearly equal to $\angle e$
- (c) $\angle i$ is more than $\angle e$, but nearly equal to $\angle r$
- (d) $\angle i$ is less than $\angle e$, but nearly equal to $\angle r$

CBSE 2016

(a)

The refractive indices of glass and water with respect to air are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If speed of light in glass is 2×10^8 m/s, find the speed of light in water.

CBSE 2016

The refractive indices of glass and water with respect to air are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If speed of light in glass is 2×10^8 m/s, find the speed of light in water.

CBSE 2016

$$\text{Refractive index of a medium} = \frac{\text{Speed of light in air}}{\text{Speed of light in the medium}}$$

$$\frac{3}{2} = \frac{\text{Speed of light in air}}{2 \times 10^8 \text{ m/s}}$$

1

$$\text{Speed of light in air} = 3 \times 10^8 \text{ m/s}$$

$$\text{Speed of light in water} = \frac{3 \times 10^8 \text{ m/s}}{4/3} = 2.25 \times 10^8 \text{ m/s}$$

1

- (a) Draw a ray diagram to show the refraction of light through a glass slab and mark angle of refraction and the lateral shift suffered by the ray of light while passing through the slab.
- (b) If the refractive index of glass for light going from air to glass is $\frac{3}{2}$, find the refractive index of air for light going from glass to air.

CBSE 2016

- (a) Draw a ray diagram to show the refraction of light through a glass slab and mark angle of refraction and the lateral shift suffered by the ray of light while passing through the slab.
- (b) If the refractive index of glass for light going from air to glass is $\frac{3}{2}$, find the refractive index of air for light going from glass to

CBSE 2016

air.

a)

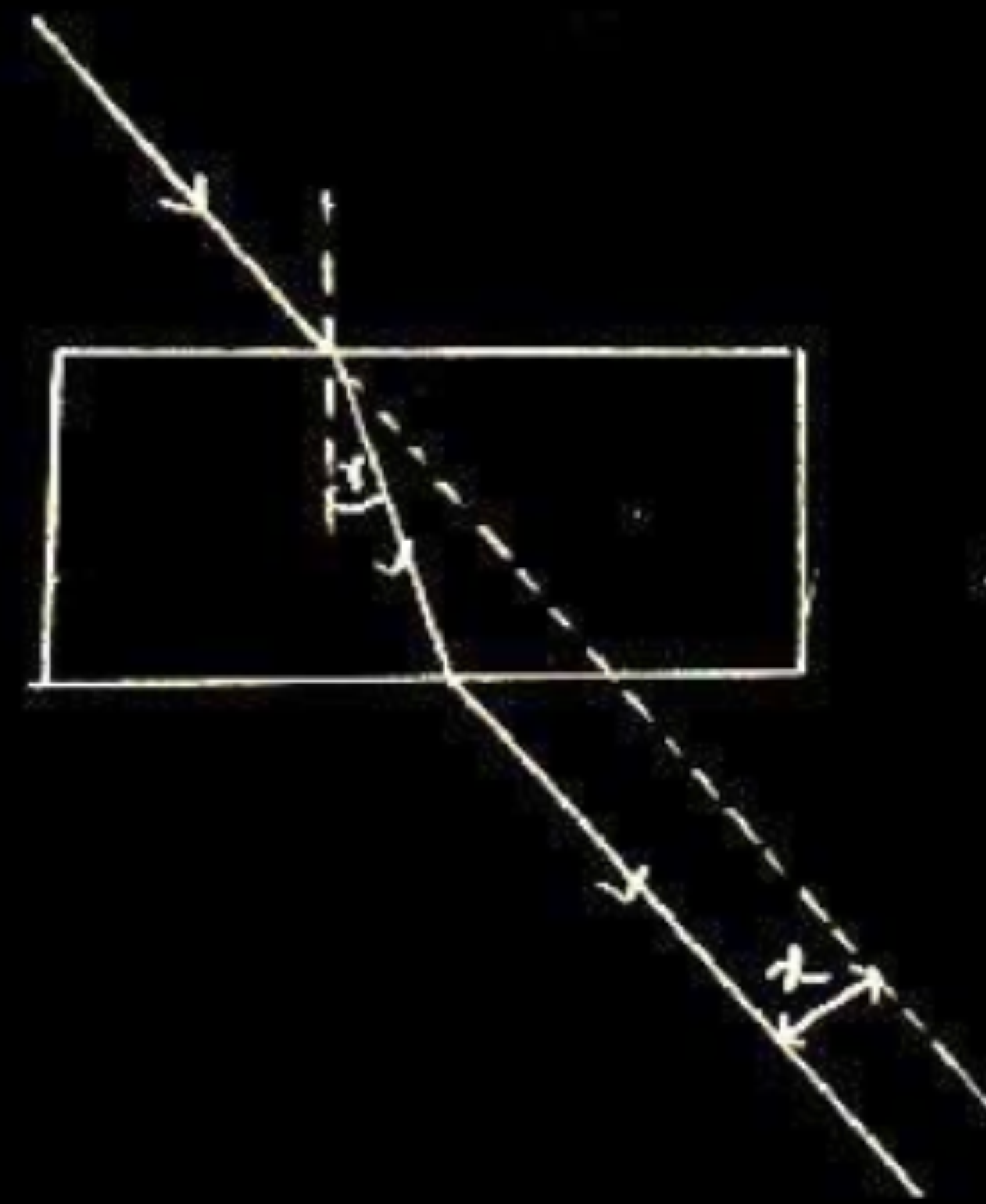


Diagram 1
Marking $\square$ r and x $\frac{1}{2}, \frac{1}{2}$

b)

$${}_a n_g = \frac{3}{2}$$

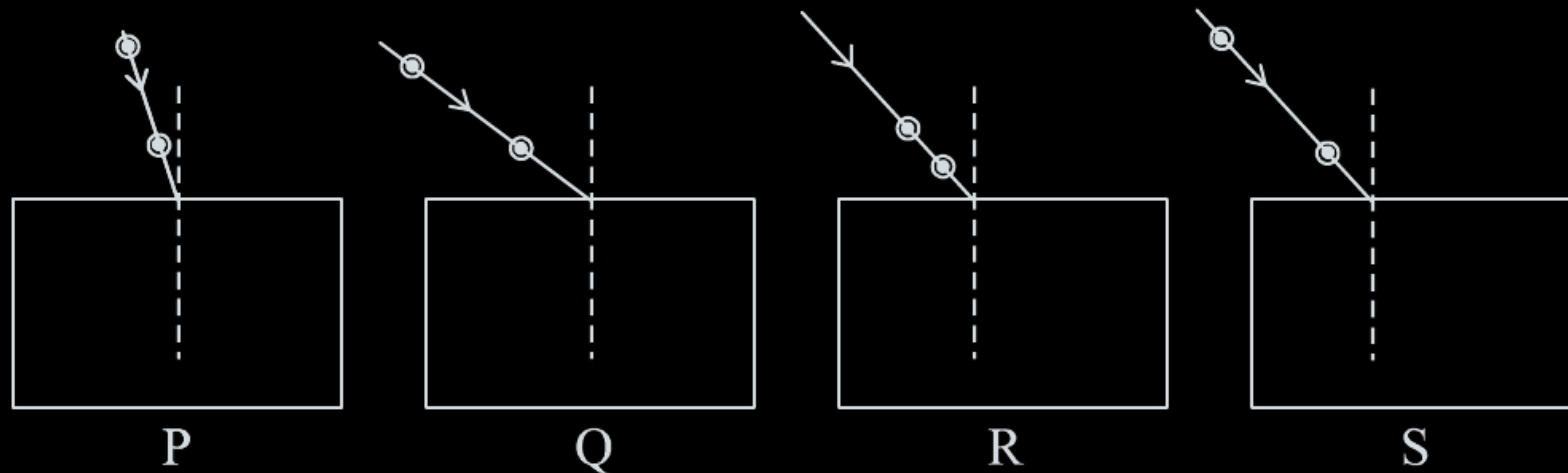
$$\square {}_g n_a = \frac{1}{{}_a n_g} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$$

Alternately,

$$\frac{c_{air}}{c_{glass}} = \frac{3}{2}$$

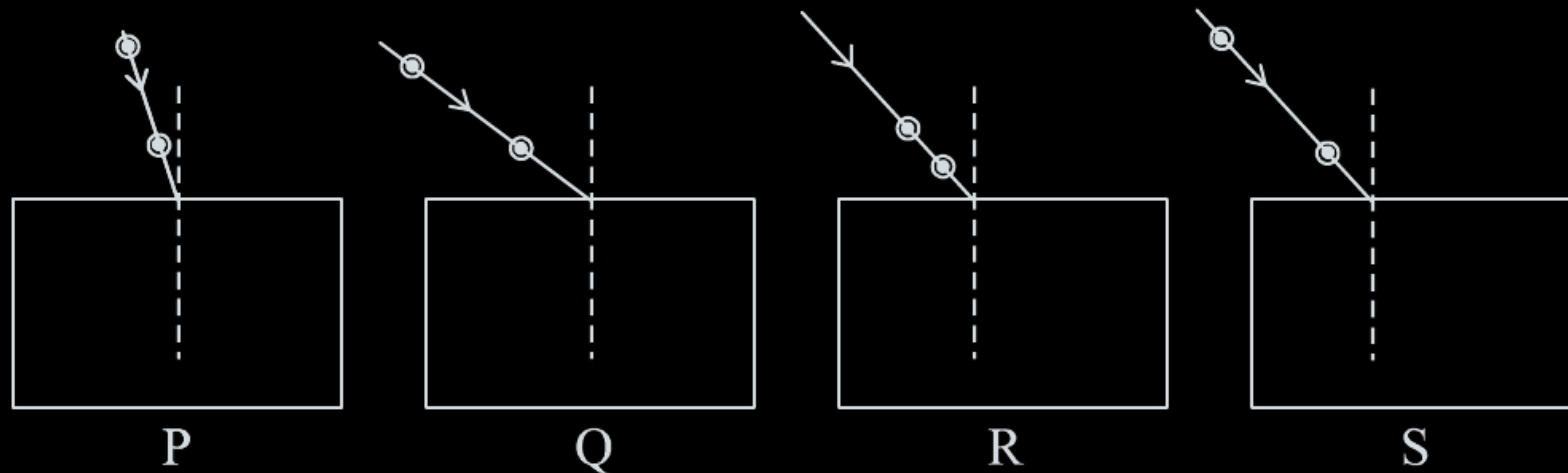
$$\square \frac{c_{glass}}{c_{air}} = \frac{2}{3}$$

Select from the following the best experimental set-up for tracing the path of a ray of light passing through a rectangular glass slab :



- (a) P
- (b) Q
- (c) R
- (d) S

Select from the following the best experimental set-up for tracing the path of a ray of light passing through a rectangular glass slab :



(a) P

(b) Q

(c) R

(d) S

CBSE 2016

(d)